

ADRC-Based Tube MPC Approach for Trajectory Tracking of Four-Mecanum-Wheel Mobile Vehicle

Jinhui Zhang , Sihang Li, Yuanqing Wu , *Senior Member, IEEE*, Ran Huang , and Huan Meng

Abstract—This article proposes a novel composite control approach based on an inner-outer-loop framework to tackle the trajectory tracking problem for the four-mecanum-wheel mobile vehicle (FMWMV) in the presence of disturbances and system constraints. To tackle the constraints of FMWMV, the modified tube model predictive control (MPC) is utilized in the outer loop. The constraints are implicitly satisfied by solving the optimal control problem. The disturbance rejection ability of MPC is enhanced with the help of the inner loop controller. **The size of the tube can be contracted by adjusting the parameters of the inner loop controller rather than being determined by the upper bound of disturbance.** It makes FMWMV more close to the optimal trajectory. To counteract the influence of disturbance, the variable-gain active disturbance rejection control method is adopted in the inner loop. The error-based variable gains are incorporated both in the tracking differentiator (TD) and the extended state observer (ESO) to improve the transient and steady-state performance. The ranges of variable-gain functions are explicitly given for the TD and ESO, respectively. Finally, the effectiveness of the proposed control approach is validated through the simulation and experimental results.

Index Terms—Active disturbance rejection control (ADRC), four-mecanum-wheel mobile vehicle (FMWMV), model predictive control (MPC).

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NOMENCLATURE

Parameters of Dynamic Model

l	Distance between geometric center and wheel center.
$2a$	Width of vehicle.
$2b$	Length of vehicle.
r	Radius of wheel.
m_{b0}	Nominal mass of vehicle body.
m_w	Mass of the identical wheels.
I	Wheel's moment of inertia about the main axis.
I_b	Vehicle's moment of inertia about the vertical axis.
ω_i	Wheel speed of the i th wheel.
τ_i	Motor torque of the i th wheel.

I. INTRODUCTION

THE four-mecanum-wheel mobile vehicle (FMWMV), which is one type of omnidirectional mobile robot [1], has gained attention due to its characteristic of direct movement in any direction. However, the application of FMWMV is less than that of nonholonomic mobile robots. The key challenge lies in improving the FMWMV's tracking performance subject to constraints and in the presence of disturbances.

Several methods have been proposed to achieve trajectory tracking for mecanum-wheel vehicle. These methods can be categorized into learning-based control methods [2], [3] or model-based control methods [4], [5]. Since the system stability under learning-based control methods cannot be rigorously proven, the most used methods are model-based ones. To apply the model-based control method, the models of the mecanum wheel and FMWMV have been researched in [6] and [7]. It has been pointed out that disturbances, such as sudden change of friction and centroid shift, greatly influence the tracking performance. To tackle the disturbances, some robust control strategies have been researched [5], [8], [9]. However, the disturbance is passively suppressed under the robust control scheme, and the accurate compensation of disturbance cannot be achieved. It is well known that the active disturbance rejection control is a useful approach to actively counteract the effects of unknown system uncertainties and disturbances, which are estimated by the well-designed extended state observer (ESO).

In the practical applications, the active disturbance rejection control (ADRC) approach may encounter the system constraints

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problem. However, system constraints must be satisfied. The main constraint is the speed saturation constraint. To overcome the input saturation problem faced with ADRC method, several researches have been done. In [10], linear and nonlinear ADRC strategies are proposed for an n -dimensional system subject to saturation constraint. The method of solving the constraint in [10] is limiting the input by $\text{sat}(\cdot)$ function. A more delicate modified $\text{sat}(\cdot)$ function is adopted in the ADRC controller in [11] to achieve the position control for a pneumatic motion platform subject to actuator saturation. However, the aforementioned methods will encounter performance degradation when the calculated input is out of the bound since the required control input cannot be achieved and the saturation constraint is not solved in an optimization-based way. Besides, there also exists speed relationship constraint [9]. The robustness of the system will be influenced if the speeds of wheels are uncoordinated [12]. However, the speed relationship constraint is not considered in ADRC approaches.

To better deal with the constraints, model predictive control (MPC) has become a mainstream for high-performance autonomous systems. It has been used in many systems, such as quadrotors [13], legged robots [14], and mobile robots [15]. MPC is also adopted to achieve the trajectory tracking for omnidirectional mobile robots. In [16], an MPC method was proposed based on an adaptive neural-fuzzy inference system, which improves the tracking accuracy. In [17], a nonlinear MPC approach was proposed to address the trajectory tracking problem. It is worth noting that the aforementioned MPC approaches overlook the disturbance of the system. To enhance the disturbance rejection ability of MPC method, some disturbance observer-based MPC methods are proposed. In [18], a disturbance observer-based data-driven MPC method is proposed. In [19], two disturbance observers are designed to estimate the unknown disturbances and the disturbances with known harmonic frequencies, respectively. By combining the disturbance observer and MPC, the influence of disturbance is alleviated. It should be pointed out that the above two methods are based on the kinematic model and the disturbances are indirectly compensated since the disturbances mainly influence the dynamic characteristic of system. Since the ADRC approach can actively compensate for the effect of disturbances based on the dynamic model and the MPC can handle the constraints, it is interesting to take advantage of these two methods to achieve the trajectory tracking of FMWMV subject to system constraints in the presence of disturbances.

In this article, a novel ADRC-based tube MPC approach under the inner-outer-loop control scheme is proposed to address the trajectory tracking for FMWMV subject to constraints in the presence of disturbance. In the outer loop, the modified tube MPC method is utilized to implicitly address the constraints and provide reference velocity for the inner loop. In the inner loop, the variable-gain ADRC method is utilized to compensate for the influence of lumped disturbances. An error-driven variable gain tracking differentiator (VGTD) is used to obtain the transient profile and derivative signal of the reference velocity. An error-driven variable gain ESO is used to observe the lumped disturbances simultaneously. A prescribed performance disturbance

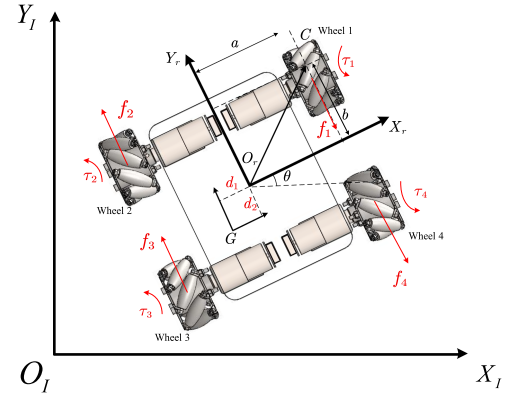


Fig. 1. Schematic diagram of the FMWMV.

rejection composite controller is used to track the reference velocity. The effectiveness of the proposed method is verified by simulation and experimental results. The contributions are summarized as follows.

- 1) The disturbance rejection ability of tube MPC is enhanced by combining it with the ADRC method. The size of the tube is no longer relevant to the upper bound of disturbance but instead can be adjusted by tuning the parameters of the inner loop controller. It makes the FMWMV more closely aligned with the optimal trajectory.
- 2) The error-driven variable-gain functions are incorporated both in tracking differentiator (TD) and ESO to improve the transient and steady-state performance. The ranges of the variable-gain functions for TD and ESO are provided, respectively. The effects of parameters in variable-gain functions are illustrated and tuning guidelines are provided.

Notations: The Minkowski addition is defined as $A \oplus B = \{a + b | a \in A, b \in B\}$. For vectors $x = [x_1, x_2, \dots, x_n]^T$ and $y = [y_1, y_2, \dots, y_n]^T$, $|x| = [|x_1|, |x_2|, \dots, |x_n|]^T$, and $x < y$ indicates that $x_i < y_i$, $i = 1, 2, \dots, n$.

II. PROBLEM FORMULATION

In this section, the kinematic model, trajectory tracking error model, and the dynamic model of the FMWMV are presented. Let $\{I\} : \{X_I, Y_I, O_I\}$ and $\{R\} : \{X_r, Y_r, O_r\}$ be the inertial coordinate frame and body coordinate frame, respectively. The parameters of FMWMV are listed in the Nomenclature and the schematic of the FMWMV is shown in Fig. 1.

A. Kinematic Model of the FMWMV

The coordinate of the FMWMV in frame $\{I\}$ is $\tilde{q} = [x_I \ y_I \ \theta]^T$. According to [20], the forward kinematic model of FMWMV in the frame $\{R\}$ can be expressed as

$$\omega \triangleq \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(a+b) \\ 1 & 1 & (a+b) \\ 1 & 1 & -(a+b) \\ 1 & -1 & (a+b) \end{bmatrix} \begin{bmatrix} v_{xr} \\ v_{yr} \\ \omega_r \end{bmatrix} \quad (1)$$

where v_{xr} and v_{yr} represent the velocities in frame $\{R\}$ and ω_r is the angular velocity. From (1), the velocity constraint of FMWMV is given as $\omega_1 + \omega_2 = \omega_3 + \omega_4$.

Define $c = \omega_1 + \omega_2 + \omega_3 + \omega_4$, $d = -\omega_1 + \omega_2 + \omega_3 - \omega_4$, $f = \frac{1}{a+b}(-\omega_1 + \omega_2 - \omega_3 + \omega_4)$, and $\tilde{u}_f = [c \ d \ f]^T$, the inverse kinematic model of FMWMV in frame $\{I\}$ is expressed as

$$\dot{\tilde{q}}(t) = \frac{r}{4} R(\theta) \tilde{u}_f(t) \quad (2)$$

where $R(\theta)$ is the rotation matrix from $\{I\}$ to $\{R\}$.

Remark 1: Assuming that an input u_f^* is gotten. Together with the constraint, the equation $A\omega = [u_f^* \ 0]^T$ is obtained, where

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -L & L & -L & L \\ 1 & 1 & -1 & -1 \end{bmatrix} \text{ with } L = \frac{1}{a+b}. \text{ Since } A \text{ has full}$$

rank, it indicates that the controller is feasible if $\|\tilde{u}_f\| \leq 4\omega_{\max}$, where $\omega_{\max}$ is the maximum speed of wheel. Thus, the constraint considered in implementation is $\mathbb{U} = \{u_f(t) \mid \|\tilde{u}_f\| \leq 4\omega_{\max}\}$.

B. Trajectory Tracking Error Model

The reference trajectory coordinate $[x_r \ y_r \ \theta_r]^T$ is generated under (2), and the tracking error is defined as

$$\tilde{q}_e(t) \triangleq \begin{bmatrix} \tilde{x}_e \\ \tilde{y}_e \\ \tilde{\theta}_e \end{bmatrix} = R(-\theta) \begin{bmatrix} x_r - x_I \\ y_r - y_I \\ \theta_r - \theta \end{bmatrix}.$$

The time derivative of $\tilde{q}_e(t)$ is given as

$$\begin{bmatrix} \dot{\tilde{x}}_e \\ \dot{\tilde{y}}_e \\ \dot{\tilde{\theta}}_e \end{bmatrix} = \begin{bmatrix} 0 & \tilde{\omega} & 0 \\ -\tilde{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{x}_e \\ \tilde{y}_e \\ \tilde{\theta}_e \end{bmatrix} + \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \\ \tilde{u}_3 \end{bmatrix} \quad (3)$$

where $\tilde{u}_1 = \frac{r}{4} \cos \tilde{\theta}_e c_r - \frac{r}{4} c$, $\tilde{u}_2 = \frac{r}{4} \sin \tilde{\theta}_e c_r + \frac{r}{4} \cos \tilde{\theta}_e d_r - \frac{r}{4} d$, $\tilde{u}_3 = \frac{r}{4} (f_r - f)$ with $c_r = \omega_{1r} + \omega_{2r} + \omega_{3r} + \omega_{4r}$, $d_r = -\omega_{1r} + \omega_{2r} + \omega_{3r} - \omega_{4r}$, and $f_r = \frac{1}{a+b}(-\omega_{1r} + \omega_{2r} - \omega_{3r} + \omega_{4r})$.

C. Dynamic Model of the FMWMV

According to [21], the dynamic model of FMWMV can be expressed in the following compact form:

$$M_0 \ddot{q} = \frac{1}{r} U + D \quad (4)$$

where $M_0 = \text{diag}\{M_1, M_2, M_3\}$, $M_1 = M_2 = m_{b0} + 4(m_w + \frac{I}{r^2})$, $M_3 = I_b + 8(m_w + \frac{I}{r^2})l^2 \sin^2(\pi/4 - \alpha)$, $\alpha = \tan^{-1} \frac{b}{a}$, $U = [u_1 \ u_2 \ u_3]^T$, and $D = [d_1 \ d_2 \ d_3]^T$.

By defining $x_1(t) = x_I(t)$ and $x_2(t) = \dot{x}_I(t)$ and extending the lump disturbance as a new state $x_3(t)$, the subsystem of x direction can be rewritten as

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= \frac{1}{M_1 r} u_1(t) + \frac{1}{M_1} d_1(t) \triangleq b_1 u_1(t) + x_3(t) \\ \dot{x}_3(t) &= h(t). \end{aligned} \quad (5)$$

For system (5), the following assumption is made.

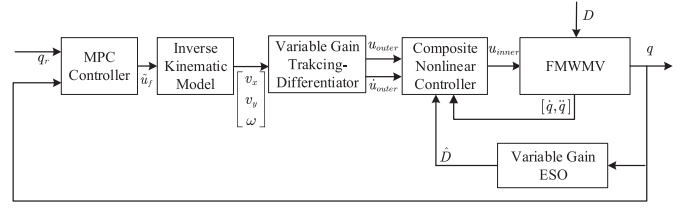


Fig. 2. Control scheme of MPC combined with ADRC.

Assumption 1: The disturbance $x_3(t)$ is assumed to be continuous and satisfies $|x_3(t)| \leq d_m$ and $|h(t)| \leq h_m$, where d_m and h_m are positive constants.

D. Control Scheme Outline

The whole control scheme is shown in Fig. 2.

q_r is the reference trajectory. $\tilde{u}_f$ is provided by the MPC algorithm. $v \triangleq [v_x \ v_y \ \omega]^T$ is provided for the inner loop by calculating the inverse kinematic model. A variable-gain TD is utilized to generate the transient profile u_{outer} and the derivative signal $\dot{u}_{\text{outer}}$. A variable-gain ESO runs simultaneously to provide the estimate $\hat{D}$ to the composite nonlinear controller. With the feedback $\dot{q}$ and $\ddot{q}$, an inner loop control input u_{inner} is calculated and implemented by the FMWMV.

III. MAIN RESULTS

A. Variable ADRC in the Inner Loop

In this section, a novel variable gain active disturbance rejection control strategy consisting of VGTD, variable gain extended state observer (VGESO), and composite disturbance rejection controller is established.

1) *Variable Gain Tracking Differentiator (VGTD):* To generate the transient profile and derivative signal of reference signal, the following TD is designed:

$$\begin{aligned} \dot{v}_1(t) &= v_2(t) \\ \dot{v}_2(t) &= -\omega_0^2 g_1(\bar{e}_1(t)) \bar{e}_1(t) - 2\omega_0 g_1(\bar{e}_1(t)) v_2(t) \end{aligned} \quad (6)$$

where ω_0 is the bandwidth parameter, $\bar{e}_1(t) = v_1(t) - v_0(t)$, $v_0(t)$ is the reference signal, $v_1(t)$ is the tracking signal, $v_2(t)$ is the derivative signal of $v_1(t)$, $g_1 = \alpha_0 + \alpha_1 e^{-\alpha_2 |\bar{e}_1(t)|}$ is the variable gain, and $\alpha_i, i = 0, 1, 2$ are positive tunable parameters. It can be verified that $g_1 \in (\alpha_0, \alpha_0 + \alpha_1]$.

Assumption 2: The reference signal $v_0(t)$ and its first-order and second-order derivative are assumed to be bounded, i.e., $|v_0(t)^{(i)}| \leq v_{\max}, i = 0, 1, 2$.

Define the tracking error of VGTD as $\bar{e}_i(t) = \frac{v_i(t) - v_0(t)^{(i-1)}}{\omega_0^{i-1}}$, $i = 1, 2$, then the tracking-error system of VGTD is given as

$$\begin{aligned} \dot{\bar{e}}_1(t) &= \omega_0 \bar{e}_2(t) \\ \dot{\bar{e}}_2(t) &= -\omega_0 g_1 \bar{e}_1(t) - 2\omega_0 g_1 \bar{e}_2(t) - \bar{u}(t) \end{aligned} \quad (7)$$

where $\bar{u}(t) = 2g_1 v_0(t)^{(1)} + v_0(t)^{(2)}/\omega_0$. With Assumption 2, there exists a positive constant $\bar{u}_{\max}$ satisfying $|\bar{u}(t)| \leq \bar{u}_{\max}$.

Rewrite the tracking error system of VGTD (7) in the following compact form:

$$\dot{\bar{e}}(t) = \omega_0 \bar{A}(\bar{e}_1(t)) \bar{e}(t) + \bar{B} \bar{u}(t) \quad (8)$$

where $\bar{e}(t) = [\bar{e}_1(t) \ \bar{e}_2(t)]^T$, $\bar{A}(\bar{e}_1(t)) = \begin{bmatrix} 0 & 1 \\ -g_1 & -2g_1 \end{bmatrix}$, and $\bar{B} = [0 \ -1]^T$.

Then, the following theorem gives the stability analysis result of the proposed VGTD.

Theorem 1: For the error system (7), if $\alpha_0 \in (\frac{\sqrt{10}+1}{\sqrt{10}+3}, \frac{\sqrt{10}-1}{\sqrt{10}-3})$ and $\alpha_0 + \alpha_1 \leq \frac{\sqrt{10}-1}{\sqrt{10}-3}$, the tracking error $\bar{e}(t)$ exponentially converges to the region Ω_1 , which is described as

$$\Omega_1 \triangleq \left\{ \bar{e}(t) \mid \|\bar{e}(t)\| \leq \frac{2(2+\sqrt{2})\|P_0 \bar{B}\| \bar{u}_{\max}}{(2-\sqrt{2})\omega_0 \kappa} \right\}$$

where κ is a positive constant.

Proof: Choose the Lyapunov function as $V_0(t) = \bar{e}(t)^T P_0 \bar{e}(t) / \omega_0$, where P_0 is a positive definite matrix satisfying $\bar{A}_0^T P_0 + P_0 \bar{A}_0 = -I_2$ with $\bar{A}_0 = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$.

Obviously, the following inequality holds:

$$\lambda_{\min}(P_0) \|\bar{e}(t)\|^2 \leq \omega_0 V_0(t) \leq \lambda_{\max}(P_0) \|\bar{e}(t)\|^2. \quad (9)$$

The derivative of $V_0(t)$ with respect to time t is given as

$$\dot{V}_0(t) = \bar{e}(t)^T \bar{A}_t \bar{e}(t) + \frac{2}{\omega_0} \bar{e}^T P_0 \bar{B} \bar{u}(t)$$

where $\bar{A}_t = \bar{A}^T P_0 + P_0 \bar{A}$, and the eigenvalues of $\bar{A}_t$ can be calculated as $\lambda_{1,2}(\bar{A}_t) = \frac{1-3g_1 \pm \sqrt{10|g_1-1|}}{2}$.

Since α_0 and α_1 satisfy the condition in Theorem 1, the following two conditions can be satisfied.

- 1) When $g_1 = 1$, $\lambda_1(\bar{A}_t) = \lambda_2(\bar{A}_t) = -1$.
- 2) When $g_1 \neq 1$, $\begin{cases} \lambda_1(\bar{A}_t) = \frac{1+\sqrt{10}}{2} - \frac{3+\sqrt{10}}{2}g_1 < 0 \\ \lambda_2(\bar{A}_t) = \frac{1-\sqrt{10}}{2} - \frac{3-\sqrt{10}}{2}g_1 < 0. \end{cases}$

It can be concluded that there exists a positive constant κ such that $\bar{A}_t \leq -\kappa I_2$, and $\dot{V}_0(t)$ satisfies

$$\dot{V}_0(t) \leq -\kappa \|\bar{e}(t)\|^2 + \frac{2\|P_0 \bar{B}\| \bar{u}_{\max}}{\omega_0} \|\bar{e}(t)\|. \quad (10)$$

With (9) and the fact that $\dot{V}_0(t) = 2\sqrt{V_0(t)} \frac{d\sqrt{V_0(t)}}{dt}$, it can be obtained from (10) that

$$\begin{aligned} \|\bar{e}(t)\| &\leq \sqrt{\frac{\omega_0 V_0(\bar{e}(t_0))}{\lambda_{\min}(P_0)}} e^{-\frac{\kappa \omega_0}{2\lambda_{\max}(P_0)}(t-t_0)} \\ &\quad + \frac{2(2+\sqrt{2})\|P_0 \bar{B}\| \bar{u}_{\max}}{(2-\sqrt{2})\omega_0 \kappa} \left(1 - e^{-\frac{\kappa \omega_0}{2\lambda_{\max}(P_0)}(t-t_0)}\right). \end{aligned} \quad (11)$$

It implies that the tracking error exponentially converges to the region Ω_1 .

2) Variable Gain Extended State Observer (VGESO): The VGESO is designed as

$$\begin{aligned} \dot{\hat{x}}_1(t) &= \hat{x}_2(t) + 3\omega_o f_1(e_1(t))e_1(t) \\ \dot{\hat{x}}_2(t) &= \hat{x}_3(t) + 3\omega_o^2 f_1(e_1(t))e_1(t) + b_1 u_1(t) \end{aligned}$$

$$\dot{\hat{x}}_3(t) = \omega_o^3 f_1(e_1(t))e_1(t) \quad (12)$$

where ω_o is the bandwidth parameter, $e_1(t) = x_1(t) - \hat{x}_1(t)$, $f_1 = \beta_1 + \beta_2 e^{-\beta_3 |e_1(t)|}$ is the nonlinear variable gain function, and β_i , $i = 1, 2, 3$, are positive tunable parameters.

Define the estimation error as $e_i(t) \triangleq \frac{x_i(t) - \hat{x}_i(t)}{\omega_o^{i-1}}$ and $e(t) \triangleq [e_1(t), e_2(t), e_3(t)]^T$ the estimation error system can be written in the following compact form:

$$\dot{e}(t) = \omega_o A(e_1(t))e(t) + B \frac{h(t)}{\omega_o^2}$$

where $A(e_1(t)) = \begin{bmatrix} -3f_1(e_1) & 1 & 0 \\ -3f_1(e_1) & 0 & 1 \\ -f_1(e_1) & 0 & 0 \end{bmatrix}$ and $B = [0 \ 0 \ 1]^T$.

Theorem 2: Consider the VGESO (12) with designed parameters $\beta_1 \in (\frac{7-2\sqrt{6}}{5}, \frac{7+2\sqrt{6}}{5})$ and $\beta_1 + \beta_2 \in (\frac{7-2\sqrt{6}}{5}, \frac{7+2\sqrt{6}}{5})$, the estimation error $e(t)$ exponentially converges to the region $\Omega_2 \triangleq \{e(t) \mid \|e(t)\| \leq \frac{2\|B^T P_1\| h_m \lambda_{\max}(P_1)}{\omega_o^3 \beta \lambda_{\min}(P_1)}\}$, where $P_1 > 0$ is the solution of Lyapunov equation $A_0^T P_1 + P_1 A_0 = -I_3$ and

$$A_0 = \begin{bmatrix} -3 & 1 & 0 \\ -3 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}.$$

Proof: The Lyapunov function is selected as $V(t) = e(t)^T P_1 e(t) / \omega_o$, and it can be obtained that

$$\lambda_{\min}(P_1) \|e(t)\| \leq \omega_o V(t) \leq \lambda_{\max}(P_1) \|e(t)\|.$$

Define $A_t = A(e_1(t))^T P_1 + P_1 A(e_1(t))$, the sequential principal minors of A_t are $D_1 = -f_1$, $D_2 = -f_1^2 + 3f_1 - 1$, and $D_3 = 1.25(f_1^2 - 2.8f_1 + 1)$. Since β_1 and β_2 satisfy the conditions given in Theorem 2, it can be obtained that D_1 and D_3 are negative and D_2 is positive. Thus, there exists a positive scalar β such that $A_t \leq -\beta I_3$.

The derivative of $V(t)$ with respect to time t is obtained as

$$\begin{aligned} \dot{V}(t) &= e(t)^T A_t e(t) + 2B^T P_1 e(t) \frac{h}{\omega_o^3} \\ &\leq -\beta \|e(t)\|^2 + 2\|B^T P_1\| \|e(t)\| \frac{h_m}{\omega_o^3}. \end{aligned}$$

Similar to (10) and (11), it can be concluded that

$$\begin{aligned} \|e(t)\| &\leq \sqrt{\frac{\omega_o V(t_0)}{\lambda_{\min}(P_1)}} e^{-\frac{\beta \omega_o}{2\lambda_{\max}(P_1)}(t-t_0)} \\ &\quad + \frac{2\|B^T P_1\| h_m \lambda_{\max}(P_1)}{\omega_o^3 \beta \lambda_{\min}(P_1)} \left(1 - e^{-\frac{\beta \omega_o}{2\lambda_{\max}(P_1)}(t-t_0)}\right). \end{aligned}$$

Thus, the estimation error $e(t)$ exponentially converges to the region Ω_2 .

3) Composite Disturbance Rejection Controller: Define the tracking error of inner loop as $z_1(t) = x_2(t) - u_{\text{outer}}(t)$. Inspired by the work in [22], the prescribed performance boundary function is given as $F_\vartheta(t) = (\vartheta_0 - \vartheta_1)e^{-t} + \vartheta_1$, where ϑ_0 and ϑ_1 are positive constants, and the funnel variable is given as $\xi(t) = \frac{z_1(t)/F_\vartheta(t)}{1-|z_1(t)|/F_\vartheta(t)}$. The derivative of $\xi(t)$ with respect to time t is obtained as

$$\dot{\xi}(t) = \frac{1}{F_\vartheta(t) - |z_1(t)|} [b_1 u_1(t) + x_3(t) - \dot{u}_{\text{outer}}(t)]$$

$$-\xi(t)(\dot{F}_\vartheta(t) - |\dot{z}_1(t)|)] .$$

The controller of inner loop is given as

$$u_1(t) = \frac{1}{b_1} [-k_x \rho(t) \xi(t) + \dot{u}_{\text{outer}} - \hat{x}_3(t) + \xi(t)(\dot{F}_\vartheta(t) - |\dot{z}_1(t)|)] \quad (13)$$

where k_x is a tunable positive constant, $\rho(t) \triangleq \frac{1}{F_\vartheta(t) - |\dot{z}_1(t)|}$.

Lemma 1 (See [23]): The stabilization of funnel variable $\xi(t)$ is sufficient to guarantee the evolution of tracking error $z_1(t)$ within the prescribed performance boundary function $F_\vartheta(t)$.

Then, the following theorem gives the stability analysis result of the inner loop system.

Theorem 3: For the inner loop system with controller (13), if the initial condition $z_1(0) < F_\vartheta(0)$ and $k_x > \frac{1}{2}$ are satisfied, the tracking error $z_1(t)$ is bounded by the prescribed performance boundary function $F_\vartheta(t)$.

Proof: The Lyapunov function is $V_{in}(t) = \frac{1}{2} \xi(t)^2$. By using the Young's inequality, the derivative of $V_{in}(t)$ with respect to time t is obtained as

$$\begin{aligned} \dot{V}_{in}(t) &= -k_x \rho(t)^2 \xi(t)^2 + \rho(t) \xi(t) e_3(t) \\ &\leq -\left(k_x - \frac{1}{2}\right) \rho(t)^2 \xi(t)^2 + \frac{\omega_0^4}{2} \|e(t)\|^2. \end{aligned} \quad (14)$$

With Theorem 2 and the fact that $V(t_0)$ is bounded, there exists a positive constant $e_{\max}$ such that $\|e(t)\|^2 \leq e_{\max}$. Then, (14) can be further induced as

$$\dot{V}_{in}(t) \leq -\left(k_x - \frac{1}{2}\right) \rho(t)^2 \xi(t)^2 + \frac{\omega_0^4 e_{\max}}{2}. \quad (15)$$

Similar to (10) and (11), it can be obtained from (15) that

$$\dot{V}_{in}(t) \leq V_{in}(t_0) e^{-\varrho t} + \frac{\sigma}{\varrho} (1 - e^{-\varrho t}) \quad (16)$$

where $\varrho = k_x - 1/2$ and $\sigma = \omega_0^4 e_{\max}/2$.

Then, one can obtain that $\xi(t)$ is bounded and converges to a bounded region. According to the Lemma 1, the tracking error of inner loop $z_1(t)$ can be retained in the prescribed performance boundary function $F_\vartheta(t)$.

B. Tube MPC in the Outer Loop

In this section, a tube MPC strategy is established to generate the desired input for inner loop.

The nominal kinematic model (2) is adopted to predict the trajectory and the cost function is given as

$$J = \int_{t_k}^{t_k+T} f(\tilde{q}_e(\tau|t_k), \tilde{u}_f(\tau|t_k)) d\tau + g(\tilde{q}_e(t_k+T|t_k))$$

where $f(\tilde{q}_e(\tau|t_k), \tilde{u}_f(\tau|t_k)) = \|\tilde{q}_e(\tau|t_k)\|_Q^2 + \|\tilde{u}_f(\tau|t_k)\|_P^2$ with $Q = \text{diag}\{q_1, q_2, q_3\}$, $P = \text{diag}\{p_1, p_2, p_3\}$, and $q_i > 0$, $p_i > 0$, $i = 1, 2, 3$, $g(\tilde{q}_e(t_k+T|t_k)) = \frac{1}{2} \|\tilde{q}_e(t_k+T|t_k)\|^2$, t_k is the sampling time, and T is the prediction horizon.

Based on (2), the disturbed kinematic model is written as $\dot{q} = \frac{r}{4} R(\theta) u_f + d_q$, where $d_q = [z_1, z_2, z_3]^T$. It can be verified that

$\|d_q\| \leq \sqrt{3} F_\vartheta(t)$ by noticing that $|z_i(t)| \leq F_\vartheta(t)$, $i = 1, 2, 3$. Then, the disturbed tracking error system is written as

$$\dot{q}_e = \begin{bmatrix} 0 & \tilde{\omega} & 0 \\ -\tilde{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} q_e + \begin{bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \\ \tilde{u}_3 \end{bmatrix} - d_q. \quad (17)$$

Lemma 2 (See [24]): System (17) is input-to-state-stable (ISS) if there exist a class $\mathcal{KL}$ function $\alpha(\cdot, \cdot) : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ and a class $\mathcal{K}$ function $\beta(\cdot)$ satisfy the following:

$$\|q_e(t)\| \leq \alpha(\|q_e(t_0)\|, t) + \beta(z) \quad \forall t \geq 0.$$

Lemma 3: For system (3), the terminal set can be written as $\Omega = \{\tilde{q}_e | \sqrt{k_1^2 \tilde{x}_e^2 + k_2^2 \tilde{y}_e^2 + k_3^2 \tilde{\theta}_e^2} \leq r(1 - \lambda) \omega_{\max}\}$, where $\lambda \in (0, 1]$. Let $u_r = [c_r, d_r, f_r]^T$ be the inverse kinematic input of virtual leader. If $\|u_r\| \leq \frac{4\lambda\omega_{\max}}{\|R_v(\tilde{\theta}_e)\|}$, $p_i q_i < \frac{1}{4}$, and $k_i \in (\frac{1-\sqrt{1-4p_i q_i}}{2p_i}, \frac{1+\sqrt{1-4p_i q_i}}{2p_i})$, $i = 1, 2, 3$, then the terminal controller can be rewritten as $\tilde{u}_f^\kappa = \frac{4}{r} K_t \tilde{q}_e + R_v(\tilde{\theta}_e) u_r$, where

$$K_t = \text{diag}\{k_1, k_2, k_3\} \text{ and } R_v(\tilde{\theta}_e) = \begin{bmatrix} \cos \tilde{\theta}_e & 0 & 0 \\ \sin \tilde{\theta}_e & \cos \tilde{\theta}_e & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Proof: It can be verified that $\|R_v(\tilde{\theta}_e)\| \leq 1 \forall \tilde{\theta}_e \in (-\pi, \pi]$. If $\tilde{q}_e \in \Omega$, then the Euclidean norm of $\tilde{u}_f^\kappa$ satisfies

$$\begin{aligned} \|\tilde{u}_f^\kappa\| &\leq \frac{4}{r} \|K_t \tilde{q}_e\| + \|R_v(\tilde{\theta}_e)\| \|u_r\| \\ &\leq 4(1 - \lambda) \omega_{\max} + 4\lambda \omega_{\max} = 4\omega_{\max} \end{aligned}$$

which indicates that $\tilde{u}_f^\kappa \in \mathbb{U}$.

When the states are in the terminal set, the Lyapunov function is chosen as $g(\tilde{q}_e(\tau|t_k))$. By applying the terminal controller, the derivative of $g(\tilde{q}_e(\tau|t_k))$ is obtained as

$$\dot{g} = -\left(k_1 \tilde{x}_e^2 + k_2 \tilde{y}_e^2 + k_3 \tilde{\theta}_e^2\right) < 0$$

which means that terminal set Ω is an invariant set.

Since p_i , q_i , and k_i satisfy the conditions in Lemma 3, the following inequality is obtained:

$$\begin{aligned} &f(\tilde{q}_e(t|t_k), \tilde{u}(t|t_k)) + \dot{g}(\tilde{q}_e(t|t_k)) \\ &= (p_1 k_1^2 - k_1 + q_1) \tilde{x}_e^2 + (p_2 k_2^2 - k_2 + q_2) \tilde{y}_e^2 \\ &\quad + (p_3 k_3^2 - k_3 + q_3) \tilde{\theta}_e^2 < 0 \end{aligned} \quad (18)$$

which means $\tilde{u}_f^\kappa$ guarantees the convergence of the system.

Then, the tube MPC optimal control problem (OCP) is formulated as

$$\min_{\tilde{u}_f(\tau|t_k)} J(\tilde{q}_e(t_k), \tilde{u}_f(t_k)) \quad (19a)$$

$$s.t. \quad \dot{\tilde{q}}(\tau|t_k) = \frac{r}{4} R(\tilde{\theta}) \tilde{u}_f(\tau|t_k) \quad (19b)$$

$$q(t_k) \in \tilde{q}^*(t_k|t_k) \oplus \mathbb{Q}_{f_e} \quad (19c)$$

$$\tilde{u}_f(\tau|t_k) \in \mathbb{U}_c \quad (19d)$$

$$\tilde{q}_e(t_k + T|t_k) \in \Omega_c \quad (19e)$$

where

$$\begin{aligned} \mathbb{Q}_{fe} &= \{q_{fe}(t) \mid |q_{fe}(t)| \leq [\bar{q}_{tube_1}, \bar{q}_{tube_2}, \bar{q}_{tube_3}]^T\} \\ \bar{q}_{tube_i} &= -\frac{\vartheta_{0i} - \vartheta_{1i}}{k_{f_i} + 1} \left| \frac{e^{-t_{fe}} - e^{k_{f_i} t_{fe}}}{k_{f_i} + 1} \right| - \frac{\vartheta_{1i}}{k_{f_i}}, i = 1, 2, 3 \\ t_{fe} &= \frac{\ln(-1/k_{f_i})}{1 + k_{f_i}}, \lambda = \frac{r}{16\omega_{\max}} \|K_f \bar{q}_{tube}\| \\ \mathbb{U}_c &= \{\tilde{u}_f(\tau|t_k) \mid \|\tilde{u}_f(\tau|t_k)\| \leq 4\omega_{\max}(1 - \lambda)\} \\ \Omega_c &= \left\{ \tilde{q}_e(t) \mid \sqrt{k_1^2 \tilde{x}_e^2 + k_2^2 \tilde{y}_e^2 + k_3^2 \tilde{\theta}_e^2} \leq r(1 - \lambda)\omega_{\max} \right\} \end{aligned}$$

where $\mathbb{Q}_{fe}$ is the tube. $\mathbb{U}_c$ is the contracted control input constraint. Ω_c is the terminal set.

Define the error between the actual trajectory q and the optimal trajectory $\tilde{q}^*$ as $q_{fe} = q - \tilde{q}^*$, and the derivative of q_{fe} can be obtained as

$$\dot{q}_{fe} = \frac{r}{4} (R(\theta)u_f - R(\tilde{\theta}^*)\tilde{u}_f) + d_q \quad (20)$$

where $u_f(t), t \in (t_k, t_{k+1}]$, is the controller in practical implementation, which is given as

$$u_f(t) = R^{-1}(\theta) \left(R(\tilde{\theta}^*)\tilde{u}_f(t|t_k) + \frac{4}{r} K_f q_{fe}(t) \right) \quad (21)$$

where $K_f = \text{diag}\{k_{f_1}, k_{f_2}, k_{f_3}\}$ and $k_{f_i} < 0, i = 1, 2, 3$, are constants.

The following theorem shows how to construct the tube and the contracted controller constraint.

Theorem 4: For disturbed tracking error system (17), if ϑ_i and K_f are selected such that $\lambda < 1$ and the parameters of inner loop controller satisfies the conditions in Theorem 3, the controller (21) makes the actual trajectory $q(t)$ stay in $\mathbb{Q} = \tilde{q}^* \oplus \mathbb{Q}_{fe}$ and the controller (21) satisfies $u_f(t) \in \mathbb{U}$, if $\tilde{u}_f \in \mathbb{U}_c$.

Proof: Substituting (21) into (20), it can be obtained that

$$\dot{q}_{fe} = K_f q_{fe} + d_q. \quad (22)$$

Solving (22), it can be obtained that

$$q_{fe}(t) = e^{K_f t} q_{fe}(0) + \int_0^t e^{K_f(t-\tau)} d_q(\tau) d\tau.$$

Since the initial state $q(t_k) \in \tilde{q}^*(t_k|t_k) \oplus \mathbb{Q}_{fe}$, and $|z_i(t)| \leq F_\vartheta(t)$, if $k_{f_i} \neq -1$, the following inequality can be obtained:

$$|q_{fe}(t)| \leq \begin{bmatrix} -\frac{\vartheta_{01}-\vartheta_{11}}{k_{f_1}+1} (e^{-t} - e^{k_{f_1} t}) - \frac{\vartheta_{11}}{k_{f_1}} (1 - e^{k_{f_1} t}) \\ -\frac{\vartheta_{02}-\vartheta_{12}}{k_{f_2}+1} (e^{-t} - e^{k_{f_2} t}) - \frac{\vartheta_{12}}{k_{f_2}} (1 - e^{k_{f_2} t}) \\ -\frac{\vartheta_{03}-\vartheta_{13}}{k_{f_3}+1} (e^{-t} - e^{k_{f_3} t}) - \frac{\vartheta_{13}}{k_{f_3}} (1 - e^{k_{f_3} t}) \end{bmatrix}. \quad (23)$$

It can be verified that $f_{e_i}(t) \triangleq \frac{\vartheta_{0i}-\vartheta_{1i}}{k_{f_i}+1} (e^{-t} - e^{k_{f_i} t}) < 0$, $i = 1, 2, 3$, and $f_{e_i}(t)$ reaches the minimum value $\bar{f}_{e_i} = \frac{\vartheta_{0i}-\vartheta_{1i}}{k_{f_i}+1} (e^{-t_{fe_i}} - e^{k_{f_i} t_{fe_i}})$ at instant t_{fe_i} . Then, by noticing that $e^{k_{f_i} t} \in (0, 1]$, it can be derived from (23) that

$$q_{fe} \in \mathbb{Q}_{fe} = \left\{ q_{fe} \mid |q_{fe}| \leq \bar{q}_{tube} \triangleq \begin{bmatrix} -\bar{f}_{e_1} - \frac{\vartheta_{11}}{k_{f_1}} \\ -\bar{f}_{e_2} - \frac{\vartheta_{12}}{k_{f_2}} \\ -\bar{f}_{e_3} - \frac{\vartheta_{13}}{k_{f_3}} \end{bmatrix} \right\}. \quad (24)$$

From the definition of $q_{fe}(t)$, it can be obtained that $q(t) = \tilde{q}^*(t) + q_{fe}(t) \in \mathbb{Q}$.

Define $u_f^a = R(\theta)u_f$ and $\tilde{u}_f^{a*} = R(\tilde{\theta}^*)\tilde{u}_f$, then it can be obtained that $u_f^a = \tilde{u}_f^a + \frac{4}{r} R(\theta) K_f q_{fe}(t)$. Since u_f and $\tilde{u}_f^a$ are, respectively, affine transformations of u_f and $\tilde{u}_f$, the constraints of u_f^a and $\tilde{u}_f^a$ are the same as that of u_f and $\tilde{u}_f$, respectively. The proof of $u_f(t) \in \mathbb{U}$, if $\tilde{u}_f \in \mathbb{U}_c$, is equivalent to the proof of $u_f^a(t) \in \mathbb{U}$, if $\tilde{u}_f^a \in \mathbb{U}_c \forall \theta \in (-\pi, \pi]$, and $\forall \tilde{\theta}^* \in (-\pi, \pi]$.

With the fact that (21d) holds, the Euclidean norm of u_f^a is obtained that

$$\begin{aligned} \|u_f^a\| &\leq \|\tilde{u}_f^{a*}\| + \frac{r}{4} \|K_f q_{fe}\| \\ &\leq 4\omega_{\max} - \frac{r}{4} \|K_f \bar{q}_{tube}\| + \frac{r}{4} \|K_f \bar{q}_{tube}\| = 4\omega_{\max}. \end{aligned}$$

It further indicates that $\|\tilde{u}_f^{a*}\| \leq 4\omega_{\max}(1 - \lambda)$, where $\lambda = \frac{r}{16\omega_{\max}} \|K_f \bar{q}_{tube}\|$.

It can be concluded that $u_f^a \in \mathbb{U}$, if $\tilde{u}_f^a \in \mathbb{U}_c$, for $\forall \theta \in (-\pi, \pi]$, and $\forall \tilde{\theta}^* \in (-\pi, \pi]$.

Remark 2: It can be seen from (24) that the size of tube can be adjusted by tuning the parameters ϑ_0, ϑ_1 , and k_{f_i} . A larger $\vartheta_0 - \vartheta_1$ and larger $|k_{f_i}|$ lead to smaller tube.

Theorem 5: The OCP (19) is feasible at any successive time instant $t > t_0$, if it is feasible at initial time t_0 , and the disturbed tracking error system (17) is ISS.

Proof: Assume that an input sequence $\tilde{u}_f(t_k)$ and optimal trajectory $\tilde{q}^*(t_k)$ are obtained at instant t_k by solving (19), and $\tilde{q}^*(t_k + T|t_k) \in \Omega_c$. Then, at time instant t_{k+1} , the actual trajectory $q(t_{k+1}) \in \tilde{q}^*(t_{k+1}|t_k) \oplus \mathbb{Q}_{fe}$, which means that $\tilde{q}^*(t_{k+1}|t_k)$ is a feasible initial state for (19) at t_{k+1} . Then, the feasible solution for (19) at t_{k+1} can be constructed as

$$\tilde{u}_f(t_{k+1}) = \begin{cases} \tilde{u}_f(\tau|t_k), & \tau \in [t_{k+1}, t_k + T) \\ \tilde{u}_f^*(\tau), & \tau \in [t_k + T, t_{k+1} + T). \end{cases}$$

Since the terminal set Ω_c is an invariant set under the terminal controller $\tilde{u}_f^*$, it can be obtained that $\tilde{q}_e(t_{k+1} + T) \in \Omega_c$. Thus, the recursive feasibility is proven according to mathematical induction.

For nominal tracking error system (3), choose a Lyapunov function as $V = J(\tilde{q}_e^*(t_k), \tilde{u}_f(t_k))$. Between the two successive instants t_k and t_{k+1} , the variation of V is denoted as

$$\begin{aligned} \Delta V &= \int_{t_k+T}^{t_{k+1}+T} \|\tilde{q}_e(\tau|t_{k+1})\|_Q^2 + \|\tilde{u}_f(\tau|t_{k+1})\|_P^2 d\tau \\ &\quad - \int_{t_k}^{t_{k+1}} \|\tilde{q}_e^*(\tau|t_k)\|_Q^2 + \|\tilde{u}_f^*(\tau|t_k)\|_P^2 d\tau \\ &\quad + \frac{1}{2} \|\tilde{q}_e(t_{k+1} + T|t_{k+1})\| - \frac{1}{2} \|\tilde{q}_e^*(t_k + T|t_k)\|. \end{aligned}$$

Calculating the integral of (18) from $t_k + T$ to $t_{k+1} + T$, it can be obtained that $\Delta V < 0$. Then, the states of nominal tracking error system (3) asymptotically converge to zero. Thus, there exists a $\mathcal{KL}$ class function $\alpha(\cdot, \cdot)$ such that $\tilde{q}_e(t) \leq \alpha(\|\tilde{q}_e(t_0)\|, t) \forall t > t_0$. Noticing that $q_{fe} \in \mathbb{Q}_{fe}$, there exists a $\mathcal{K}$ class function $\gamma(\cdot)$ such that $\|q_{fe}(t)\| \leq \gamma(z) \forall t > t_0$, where $z = \max\{\bar{q}_{tube_i}\}$.

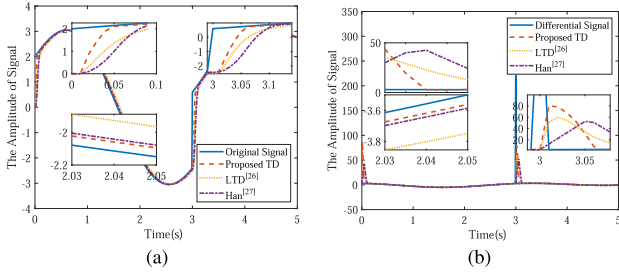


Fig. 3. Tracking error of different TDs. (a) Tracking of $v_0(t)$ with TDs. (b) Estimation of $\dot{v}_0(t)$ with TDs.

TABLE I
PARAMETERS IN DIFFERENT TDs

VGTD (6)	$\omega_1 = 40$	$\alpha_0 = \frac{6(\sqrt{10}+1)}{5(\sqrt{10}+3)}$	$\alpha_1 = \frac{4\sqrt{10}}{5}$	$\alpha_2 = 0.5$
LTD [25]	$\omega_0 = 40$			
Han [26]	$r = 1000$	$h = 0.01$		

According to the definition of q_{fe} , it can be obtained that $q_e = q_e^* - R(-\theta)q_{fe}$. With the fact that $q_{fe}(t_0) = 0$, the inequality $\|q_e(t)\| \leq \alpha(\|q_e(t_0)\|, t) + \gamma(z)$ can be obtained. Thus, the disturbed tracking error system (17) is ISS.

IV. NUMERICAL SIMULATIONS AND EXPERIMENTS

A. Comparative Simulation Results Among Different TDs

In this section, the comparative simulations between the proposed VGTD, linear TD (LTD) [25], and nonlinear TD [26] are carried out. The reference signal is given as $v_0(t) = \begin{cases} 1.5 \sin(2t) + 2 \cos(t), & t \leq 3s \\ 1.5 \sin(2t) + 2 \cos(t) + 3, & t > 3s. \end{cases}$

The simulation results are shown in Fig. 3 and the parameters are given in Table I.

It can be seen that the proposed VGTD outperforms nonlinear TD and LTD considering the transient process and steady-state error. Although the peak phenomena also appear in the respect to differential signal $v_2(t)$, the peak value is similar with that of LTD, and decreases much more faster than LTD.

Remark 3: The parameter tuning follows the following steps.

- 1) Ignore the function g_1 and tune the bandwidth parameter ω_0 according to the procedure provided in [25].
- 2) Add the nonlinear function and set the parameter $\alpha_2 = 1$.
- 3) Determine the value of α_0 and α_1 under the constraints given in Theorem 1.
- 4) Reduce α_2 and increase α_0 and α_1 to shorten the convergence time. Increase α_0 and α_1 to get a small steady-state error. If there exists oscillation, then the value of α_0 and α_1 should be reduced.
- 5) Go back to step 1) if a satisfied performance cannot be obtained.

B. Comparative Simulation Results Among Different ESOs

In this section, the comparative results are given to verify the superiority of proposed VGESO (12) by taking a comparison

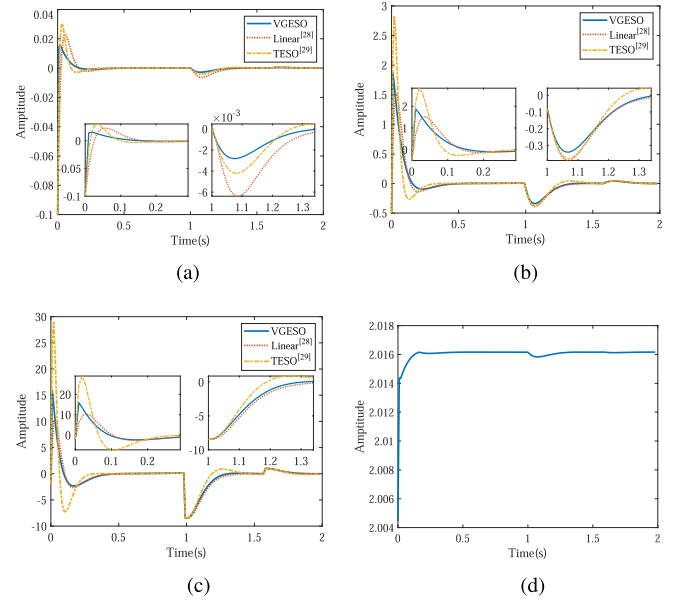


Fig. 4. Tracking error of different ESOs. (a) Estimation error of $x_{b1}(t)$. (b) Estimation error of $x_{b2}(t)$. (c) Estimation error of $d(t)$. (d) Variable gain of proposed VGESO.

TABLE II
PARAMETERS IN DIFFERENT ESOs

VGESO (12)	$\omega_0 = 30$	$\beta_1 = \frac{14-4\sqrt{6}}{5}$	$\beta_2 = \frac{12\sqrt{6}}{25}$	$\beta_3 = 10$
LESO [27]	$\omega_0 = 30$			
TESO [28]	$\omega = 30$	$\rho_\infty = 0.35$		

between the proposed ESO and the one proposed in [27] and time-varying ESO (TESO) proposed in [28].

A second-order system to be observed is described as $\begin{cases} \dot{x}_{b1}(t) = x_{b2}(t), \\ \dot{x}_{b2}(t) = b_0 u(t) + d(t), \end{cases}$ where $b_0 = 1$, $u(t) = \cos(t)$, and the disturbance is given as $d(t) = \begin{cases} \sin(t) + 2 \cos(t) & t \leq 1s \\ 5 + 0.2 \sin(\cos(t)) & t > 1s \end{cases}$.

The simulation results are shown in Fig. 4 and the observer parameters are given in Table II.

It can be seen that the proposed VGESO (12) outperforms the comparative ESOs considering both the transient and steady-state performances. Especially, at the initial transient process, the proposed VGESO possesses smaller overshoot and faster convergence speed.

Remark 4: The parameter tuning follows the following steps.

- 1) Ignore the function f_1 and tune the bandwidth parameter ω_o according to the procedure provided in [27].
- 2) Add the nonlinear function and set the parameter $\beta_3 = 1$.
- 3) Determine the value of β_1 and β_2 under the constraints in Theorem 2.
- 4) Increase β_1 and β_2 to shorten the convergence time and decrease steady-state error. If there exists oscillation, then the value of β_1 , β_2 , and β_3 should be decreased.
- 5) Go back to step 1) if a satisfied performance cannot be obtained.

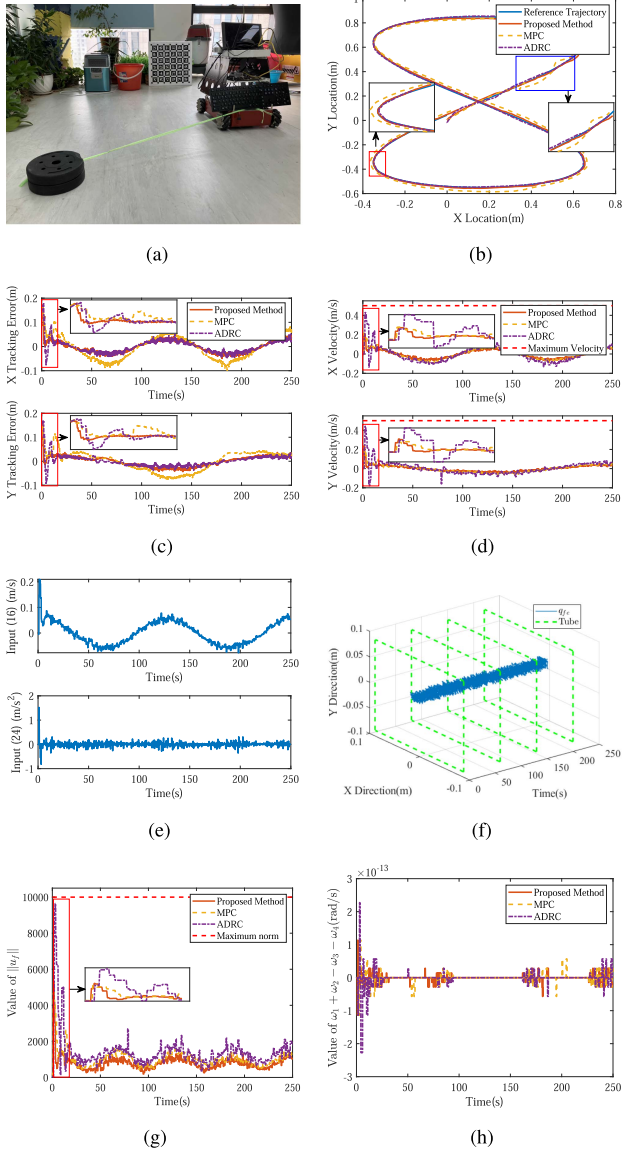


Fig. 5. Tracking performance of FMWMV under different methods. (a) FMWMV in experiment. (b) Overall tracking result. (c) Tracking error of FMWMV. (d) FMWMV's Velocity. (e) Control input (13) and (21). (f) Evolution of q_{fe} . (g) Input constraint of $\|u_f\|$. (h) Validation of Remark 1.

C. Experimental Trajectory Tracking Control Results

In the experiment, the proposed composite control approach is compared with tube MPC method and ADRC control method. The FMWMV is shown in Fig. 5(a). An eight-shaped trajectory with initial position at (0.15 0.15) is tracked while the FMWMV starts at the original point (0 0). A rope with a 6 kg load is bound behind the FMWMV and the rope is loose at the initial time. A robot operation system (ROS) runs on the vehicle and the laptop is connected to ROS via WiFi. The laptop computes the control input and transmits it to the vehicle. The parameters of different controller are listed in Table III. The maximum velocity of each direction is set to be 0.5 m/s in experiment. By searching the user's manual of the vehicle, the $\|\tilde{u}_f\|$ should

TABLE III
CONTROLLER PARAMETERS

Proposed controller	$Q = 0.6I_3$	$P = 0.4I_3$	$K_f = -2I_3$
	$\vartheta_{0i} = 0.3$	$\vartheta_{1i} = 0.1$	$(i = 1, 2, 3)$
	$k_x = 1$	$k_y = 1$	$k_w = 1$
Tube MPC controller	$Q = 0.6I_3$	$P = 0.4I_3$	$K_f = -2I_3$
ADRC	$\alpha_1 = 0.5$	$\alpha_2 = 0.25$	$\delta = 0.1$
	$\beta_1 = 2.3$	$\beta_2 = 1$	

be set at 10 000. The results can be concluded from the following aspects.

- 1) Fig. 5(b) and (c) demonstrates the effectiveness of enhancing disturbance rejection ability. The right-hand side enlarged figure in Fig. 5(b) shows the tracking result when the rope is first tightened. It can be seen that the FMWMV deviates less from the desired trajectory under the control of the proposed method. Meanwhile, there appears an oscillation in tracking under the control of the compared tube MPC method. It can be seen from the enlarged figures in Fig. 5(c) that the tracking error under the control of the proposed method is basically unchanged, while the tracking error under the control of the compared tube MPC method varies more drastically. Fig. 5(c) shows the tracking error during the whole process. It can be seen that the tracking error of FMWMV under the control of the proposed method is smaller than that under the control of the compared tube MPC method. Thus, the proposed method outperforms the compared tube MPC method from the aspect of the transient process and the steady-state error.
- 2) To verify the proposed method's effectiveness concerning the aspect of solving velocity constraints, we carry out the experiments by comparing the velocity of FMWMV on the premise of completing the tracking. It can be seen from Fig. 5(d) that the velocity of FMWMV under the control of ADRC is larger than that under the control of the proposed method. It indicates that the velocity constraints are more likely violated under the control of the ADRC method.
- 3) The correctness of Remark 1 is verified if the value of $\omega_1 + \omega_2 - \omega_3 - \omega_4$ equals zero. It can be seen from Fig. 5(h) that the value basically remains zeros while the nonzero instance is mainly caused by the rounding-off error. Thus, it indicates the correctness of Remark 1.

V. CONCLUSION AND FUTURE WORKS

In this article, a combination of the variable-gain ADRC and tube MPC is proposed to address the trajectory tracking for FMWMV in the presence of disturbances and system constraints. An inner-outer-loop scheme is adopted. Tube MPC

is utilized in the outer loop to implicitly address the velocity constraint and the size of the tube can be adjusted by tuning the parameters, which makes the real trajectory more closer to the optimal trajectory. ADRC is utilized in the inner loop to counteract the influence of disturbance, which enhances the disturbance rejection ability of MPC. The error-driven variable gain function is introduced in ADRC to improve the transient performance. There also remain some unexplored directions, such as how to adopt a dynamically adjusted tube rather than a rigid tube and how to mathematically analyze the dynamic performance of the ADRC method.

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